

This post will cover the strengths and weaknesses of CNN-based classification versus a traditional machine-learning classifier using manually coded features. It will focus on object recognition, in this instance a hypothetical fruit farm looking to automatically detect ripe fruit.

A Convolutional Neural Network (CNN) is widely viewed as well suited to classification and recognition tasks (Chollet, 2021; Asritha et al., 2023). As a deep-learning model, it is highly versatile. For example, it can learn to distinguish between ripe, unripe and over-ripe fruit (Karegowda, U and Sheela, 2024), even without manually programmed parameters such as colour histograms or edge detection (Ghosh and Singh, 2026).

It also produces more accurate results when trained with complex environmental variations, such as background, lighting and image quality. Furthermore, the same model could be used to identify additional characteristics, such as fruit pests, damaged items or foreign objects, without additional, manually coded parameters (Bishshash et al., 2024).

A drawback of CNNs is that they are computationally expensive to run and train. They also often require a larger set of training data.

In contrast, a classifier combined with a colour histogram and edge detection algorithm is less computationally expensive to run. It may perform better with less training data. However, it would require labelled data as input. Additionally, the conditions in which the model operates must be consistent, for example, consistent background and lighting conditions.

It is worth noting that the media recently reported that companies, such as Uber, have already spent their yearly budget on AI tools and automation (BCG, 2026), yet the return on investment for businesses using AI is unclear (Curcio, E., 2025).

As such, when looking into a learning algorithm, one of the key issues facing industry at the moment is cost and return on investment. There is also the issue of the environmental impact of AI training models (Morrison et al., 2025). Whilst it is worth noting that using a pre-trained CNN could reduce the computational effort of training the model, this would not affect the running cost.

Whilst CNNs are well suited to classification and recognition tasks, they also suffer from being expensive in cost and environmental impact to train and run. In contrast, a classifier is less versatile, but may be cheaper to train and run. In the context of a single AI identifying fruit in a stable environment, such as a production line, a classifier may be the better choice as it is less computationally expensive to implement. However, if accuracy and versatility are more important, then CNNs would be the better option.

Asritha, L., Prasad, M.H.C., Sai, K.V. and Jayaraj, M. (2023) 'Fruit Ripeness Detection using Deep Learning', *International Journal of Scientific Research in Engineering and Management*, 7(4). Available at: <https://doi.org/10.55041/IJSREM18758>

Bishshash, P. *et al.* (2024) 'A comprehensive cotton leaf disease dataset for enhanced detection and classification', *Data in Brief*, 57, article 110913. Available at: <https://doi.org/10.1016/j.dib.2024.110913>

BCG (2026) *As AI investments surge, CEOs take the lead*. Available at: <https://www.bcg.com/publications/2026/as-ai-investments-surge-ceos-take-the-lead> (Accessed: 8 June 2026).

Curcio, E. (2025) 'Introducing LCOAI: A standardized economic metric for evaluating AI deployment costs'. Available at: <https://arxiv.org/abs/2509.02596> (Accessed: 8 June 2026).

Chollet, F. (2021) *Introduction to deep learning for computer vision*. Available at: https://deeplearningwithpython.io/chapters/chapter08_image-classification/ (Accessed: 8 June 2026).

Ghosh, D. and Singh, J.P. (2026) 'A hybrid CNN-LSTM model for accurate fruit freshness classification using deep learning', *Discover Artificial Intelligence*, 6, article 134. Available at: <https://doi.org/10.1007/s44163-025-00750-7>

Karegowda, A.G., U, H.D. and Sheela, S.J. (2024) 'Fruit Classification and Ripeness Estimation using Deep Learning Models', *2024 8th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS)*, Bengaluru, India. pp. 1–6. Available at: <https://doi.org/10.1109/CSITSS64042.2024.10817041>

Morrison, J. *et al.* (2025) 'Holistically Evaluating the Environmental Impact of Creating Language Models', *arXiv*. Available at: <https://arxiv.org/abs/2503.05804> (Accessed: 8 June 2026).